

Norway

SLEIPNER

Volve F

F-3

F-3

Plan: F-3 rev#1 STGAL 151015

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-3	North Reference:	Grid
Wellbore:	F-3	Survey Calculation Method:	Minimum Curvature
Design:	F-3 rev#1 STGAL 151015	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-3		
Well Position	+N/-S	-0.74 m	Northing:
	+E/-W	-1.30 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.7821 N
			Longitude:
			1° 53' 14.8489 E
			Water Depth:
			91.00 m

Wellbore	F-3		
Magnetics	Model Name	Sample Date	Declination
			(°)
	MNETICREFERENCE	19.10.2005	-2.55
			Dip Angle
			(°)
			71.62
			Field Strength
			(nT)
			50,281

Design	F-3 rev#1 STGAL 151015		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth:
			145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	-0.74	-1.30
			Direction
			(°)
			25.76

Survey Tool Program	Date	15.10.2015			
From	To	Survey (Wellbore)	Tool Name	Description	
(m)	(m)				
145.90	220.00	F-3 rev#1 STGAL 151015 (F-3)	Inclination Only	Inclination only surveys, vertical wells (inc < 2	
220.00	550.00	F-3 rev#1 STGAL 151015 (F-3)	Keeper, sightline	Keeper tool run in sightline mode (initialisation)	
550.00	670.00	F-3 rev#1 STGAL 151015 (F-3)	NOT_useMWD gyro, Baker I Baker Hughes' MWD gyro service (GyroTrak)		
670.00	1,332.00	F-3 rev#1 STGAL 151015 (F-3)	Magn, IFR, mag-corr, reduce Magnetic Tools (MWD, EMS) without gyro-verificatio		
1,332.00	2,454.00	F-3 rev#1 STGAL 151015 (F-3)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
2,452.00	2,978.00	F-3 rev#1 STGAL 151015 (F-3)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		
2,978.00	3,888.84	F-3 rev#1 STGAL 151015 (F-3)	Magn, IFR, mag-corr, red QC Magnetic Tools (MWD, EMS) without gyro-verificatio		

Planned Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	-0.74	-1.30	0.00	0.000	0.000	0.000	
245.90	0.00	0.00	245.90	-0.74	-1.30	0.00	0.000	0.000	0.000	
365.90	2.00	220.00	365.88	-2.35	-2.65	-2.03	0.500	0.500	0.000	
485.90	4.00	220.00	485.71	-7.16	-6.69	-8.12	0.500	0.500	0.000	
541.88	4.00	193.00	541.56	-10.56	-8.38	-11.91	1.000	0.000	-14.468	

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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,331.88	4.00	193.00	2,327.20	-132.22	-36.47	-133.69	0.000	0.000	0.000	
3,416.99	86.50	23.99	3,060.49	452.35	233.77	510.23	2.500	2.281	-4.673	
3,521.23	86.50	23.99	3,066.85	547.41	276.07	614.23	0.000	0.000	0.000	
3,523.71	86.70	23.96	3,067.00	549.67	277.08	616.70	2.500	2.476	-0.347	
3,888.84	86.70	23.96	3,088.00	882.79	425.12	981.05	0.000	0.000	0.000	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude		Longitude
- hit/miss target										
- Shape										
T1 NW	0.00	0.00	3,067.00	549.67	277.08	6,479,113.00	435,327.00	58° 26' 47.7169 N		1° 53' 31.4462 E
- plan hits target center										
- Point										
T2 NW	0.00	360.00	3,088.00	882.78	425.13	6,479,446.00	435,475.00	58° 26' 58.5601 N		1° 53' 40.2341 E
- plan hits target center										
- Point										

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name			Casing Diameter (in)	Hole Diameter (in)
220.00	220.00	30"			30.000	36.000
1,332.25	1,330.00	20"			20.000	26.000
2,453.88	2,449.00	13 5/8"			13.625	17.500
2,977.57	2,900.00	9 7/8"			9.875	12.250
3,888.84	3,088.00	7"			7.000	8.500

Checked By: _____	Approved By: _____	Date: _____
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